



Book Introduction: Traffic Flow Dynamics Data, Models and Simulation

Chapter.11:Car-Following Models Based on Driving Strategies



Car-Following Models Based on Driving Strategies

The models introduced in this chapter are derived from assumptions about real driving behavior such as keeping a “safe distance” from the leading vehicle, driving at a desired speed, or preferring accelerations to be within a comfortable range.

- Two models will be introduced: The **simplified Gipps model**, and the **Intelligent Driver Model**.
- Both models use the same input variables as the sensors of **adaptive cruise control** (ACC) systems, and produce a similar driving behavior.

Model Criteria

They are defined by an acceleration function a_{mic}
or a speed function v_{mic}

$$\dot{v}_\alpha(t) = \frac{dv_\alpha(t)}{dt} = a_{mic}(s_\alpha, v_\alpha, v_l) = \tilde{a}_{mic}(s_\alpha, v_\alpha, \Delta v_\alpha). \quad (10.3)$$

$$v_\alpha(t + \Delta t) = v_{mic}(s_\alpha(t), v_\alpha(t), v_l(t)), \quad (10.7)$$

The acceleration or speed functions encoding the driving behavior should at least model the following aspects:

1. The acceleration is a strictly decreasing function of the speed. Moreover, the vehicle accelerates towards a *desired speed* v_0 if not constrained by other vehicles or obstacles:

$$\frac{\partial a_{mic}(s, v, v_l)}{\partial v} < 0, \quad \lim_{s \rightarrow \infty} a_{mic}(s, v_0, v_l) = 0 \quad \text{for all } v_l. \quad (11.1)$$

Model Criteria

The acceleration or speed functions encoding the driving behavior should at least model the following aspects:

2. The acceleration is an increasing function of the distance s to the leading vehicle:

$$\frac{\partial a_{\text{mic}}(s, v, v_l)}{\partial s} \geq 0, \quad \lim_{s \rightarrow \infty} \frac{\partial a_{\text{mic}}(s, v, v_l)}{\partial s} = 0 \quad \text{for all } v_l. \quad (11.2)$$

The inequality becomes an equality if other vehicles or obstacles (including “virtual” obstacles such as the stopping line at a red traffic light) are outside the interaction range and therefore do not influence the driving behavior. This defines the *free-flow acceleration*

$$a_{\text{free}}(v) = \lim_{s \rightarrow \infty} a_{\text{mic}}(s, v, v_l) = \geq a_{\text{mic}}(s, v, v_l). \quad (11.3)$$

3. The acceleration is an increasing function of the speed of the leading vehicle. Together with requirement (1), this also means that the acceleration decreases (the deceleration increases) with the speed of approach to the lead vehicle (or obstacle):

$$\frac{\partial \tilde{a}_{\text{mic}}(s, v, \Delta v)}{\partial \Delta v} \leq 0 \quad \text{or} \quad \frac{\partial a_{\text{mic}}(s, v, v_l)}{\partial v_l} \geq 0, \quad \lim_{s \rightarrow \infty} \frac{\partial a_{\text{mic}}(s, v, v_l)}{\partial v_l} = 0. \quad (11.4)$$

Model Criteria

The acceleration or speed functions encoding the driving behavior should at least model the following aspects:

4. A minimum gap (bumper-to-bumper distance) s_0 to the leading vehicle is maintained (also during a standstill). However, there is no backwards movement if the gap has become smaller than s_0 by past events:

$$a_{\text{mic}}(s, 0, v_l) = 0 \quad \text{for all } v_l \geq 0, \quad s \leq s_0. \quad (11.5)$$

By virtue of relation , these requirements (or plausibility conditions) for the acceleration function naturally imply conditions for the speed function v_{mic} of models formulated in terms of coupled maps.

$$a_{\text{mic}}(s, v, v_l) = \frac{v_{\text{mic}}(s, v, v_l) - v}{\Delta t}$$

Model Criteria

- A car-following model meeting these requirements is complete in the sense that it can consistently describe all situations that may arise in single-lane traffic.

Particularly, it follows that

- (i) All vehicle interactions are of finite reach
- (ii) Following vehicles are not “dragged along”

$$a_{\text{mic}}(s, v, v_l) \leq a_{\text{mic}}(\infty, v, v_l') = a_{\text{free}}(v) \quad \text{for all } s, v, v_l, \text{ and } v_l', \quad (11.6)$$

- (iii) Exists an equilibrium speed $V_e(S)$

$$v_e'(s) \geq 0, \quad v_e(0) = 0, \quad \lim_{s \rightarrow \infty} v_e(s) = v_0. \quad (11.7)$$

- These conditions are necessary but not sufficient: More realistic constraints are needed.

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Gipps' Model

Safe Speed V_{safe}

Accidents are prevented in the model by introducing a “safe speed” $V_{safe}(s, v_l)$, which depends on the distance to and speed of the leading vehicle. It is based on the following assumptions:

1. Braking maneuvers are always executed with constant deceleration b . There is no distinction between comfortable and (physically possible) maximum deceleration.

Condition 1 implies that the braking distance that the leading vehicle needs to come to a complete stop is given by

$$\Delta x_l = \frac{v_l^2}{2b}.$$

2. There is a constant “reaction time” Δt . To come to a complete stop, the driver of the considered vehicle needs not only his or her braking distance $v^2/(2b)$, but also an additional reaction distance $v \Delta t$ travelled during the reaction time

$$\Delta x = v \Delta t + \frac{v^2}{2b}.$$

3. Even if the leading vehicle suddenly decelerates to a complete stop (worst case scenario), the distance gap to the leading vehicle should not become smaller than a minimum gap s_0

$$s - (\Delta x - \Delta x_l) \geq s_0 \implies s \geq s_0 + v \Delta t + \frac{v^2}{2b} - \frac{v_l^2}{2b}$$

Model Equation

Safe Speed V_{safe}

The speed v for which the equal sign holds (the highest possible speed) defines the “safe speed”

$$s - (\Delta x - \Delta x_l) \geq s_0 \implies s \geq s_0 + v\Delta t + \frac{v^2}{2b} - \frac{v_l^2}{2b}$$

$$v_{safe}(s, v_l) = -b\Delta t + \sqrt{b^2\Delta t^2 + v_l^2 + 2b(s - s_0)}.$$

The simplified Gipps’ model is defined as an iterated map with the “safe speed” as its main component:

$$v(t + \Delta t) = \min [v + a\Delta t, v_0, v_{safe}(s, v_l)] \quad \text{Gipps’ model.} \quad (11.11)$$

This model equation reflects the following properties:

- The simulation update time step is equal to the reaction time Δt .
- If the current speed is greater than $v_{safe} - a\Delta t$ or $v_0 - a\Delta t$, the vehicle will reach the minimum of v_0 and v_{safe} during the next time step.⁵
- Otherwise the vehicle accelerates with constant acceleration a until either the safe speed or the desired speed is reached.

Steady-State Equilibrium

The homogeneous steady state implies $v(t + \Delta t) = v_l = v$, thus

$$v = \min(v_0, v_{\text{safe}}) = \min \left(v_0, -b\Delta t + \sqrt{b^2 \Delta t^2 + v^2 + 2b(s - s_0)} \right),$$

which yields the steady-state speed-gap relation

$$v_e(s) = \max \left[0, \min \left(v_0, \frac{s - s_0}{\Delta t} \right) \right] \quad (11.12)$$

Assuming constant vehicle lengths l , the familiar “triangular” fundamental diagram

$$Q_e(\rho) = \min \left(v_0 \rho, \frac{1 - \rho l_{\text{eff}}}{\Delta t} \right), \quad (11.13)$$

where $l_{\text{eff}} = (l + s_0)$

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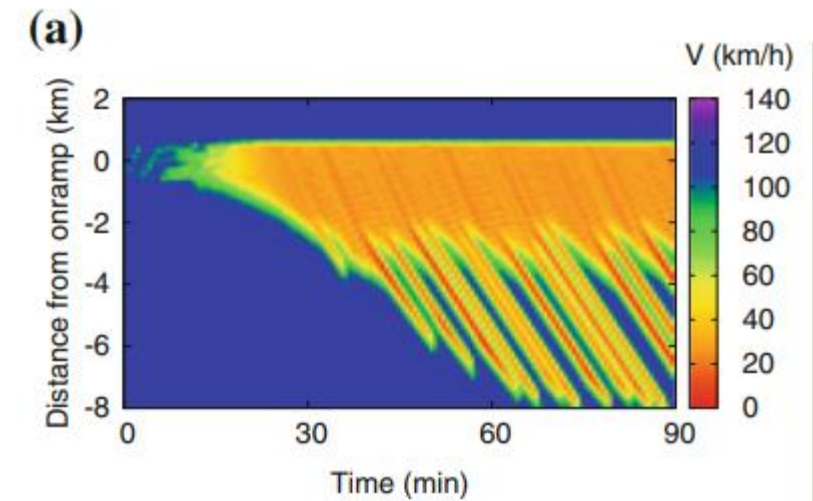
Model Characteristics

Gipps' model is transparently derived from a few basic assumptions and uses parameters that are easy to interpret and assign realistic values

Table 11.1 Parameters of the simplified Gipps' model and typical values in different scenarios

Parameter	Typical value	Typical value
	Highway	City traffic
Desired speed v_0	120 km/h	54 km/h
Adaption/reaction time Δt	1.1 s	1.1 s
Acceleration a	1.5 m/s ²	1.5 m/s ²
Deceleration b	1.0 m/s ²	1.0 m/s ²
Minimum distance s_0	3 m	2 m

Highway traffic

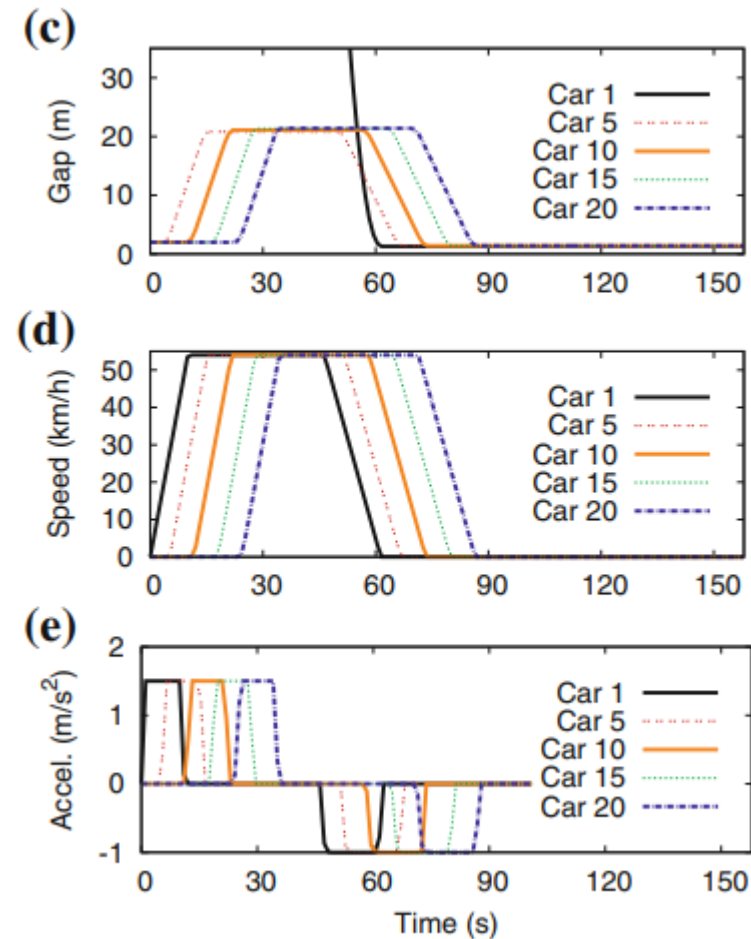


Model Characteristics

However, the acceleration time-series is unrealistic. By definition, there are only three values for the acceleration: Zero, a , and $-b$. The resulting driving behavior is excessively “robotic” and the abrupt transitions are unrealistic.

Moreover, Gipps' model does not differentiate between comfortable and maximum deceleration

1. Assuming that b denotes the maximum deceleration, the model is accident-free but every braking maneuver is performed very uncomfortably with full brakes.
2. On the other hand, when interpreting b as the comfortable deceleration and allowing for heterogeneous and/or multi-lane traffic the model possibly produces accidents if leading vehicles brake harder than b .

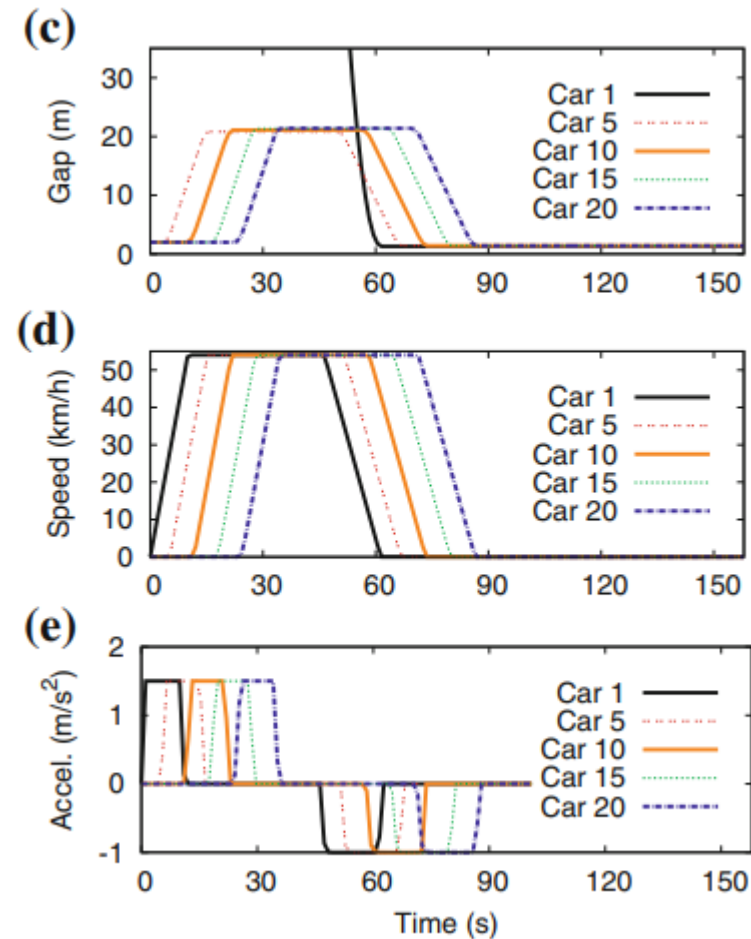


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Intelligent Driver Model

As Gipps' model, the IDM is derived from a list of basic assumptions (first-principles model). The following requirements characterize it:

1. The acceleration fulfills the general conditions (11.1)–(11.5) for a complete model.
2. The equilibrium bumper-to-bumper distance to the leading vehicle is not less than a “safe distance” $s_0 + vT$ where s_0 is a minimum (bumper-to-bumper) gap, and T the (bumper-to-bumper) time gap to the leading vehicle.
3. An *braking strategy* controls how slower vehicles (or obstacles or red traffic lights) are approached:
 - Under normal conditions, the braking maneuver is “soft”, i.e., the deceleration increases gradually to a comfortable value b , and decreases smoothly to zero just before arriving at a steady-state car-following situation or coming to a complete stop.
 - In a critical situation, the deceleration exceeds the comfortable value until the danger is averted. The remaining braking maneuver (if applicable) will be continued with the regular comfortable deceleration b .

Intelligent Driver Model

As Gipps' model, the IDM is derived from a list of basic assumptions (first-principles model). The following requirements characterize it:

4. Transitions between different driving modes (e.g., from the acceleration to the car-following mode) are smooth. In other words, the time derivative of the acceleration function, i.e., the *jerk* J , is finite at all times.⁶ This is equivalent to postulating that the acceleration function $a_{\text{mic}}(s, v, v_l)$ (or $\tilde{a}_{\text{mic}}(s, v, \Delta v)$) is continuously differentiable in all three variables. Notice that this postulate is in contrast to the action-point models such as the *Wiedemann Model* where acceleration changes are modeled as a series of discrete jumps.
5. The model should be as parsimonious as possible.

Intelligent Driver Model

The required properties are realized by the following acceleration equation:

$$\dot{v} = a \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right] \text{ IDM.} \quad (11.14)$$

The acceleration of the Intelligent Driver Model is given in the form $\tilde{a}_{\text{mic}}(s, v, \Delta v)$ and consists of two parts, one comparing the current speed v to the desired speed v_0 , and one comparing the current distance s to the desired distance s^* . The desired distance

$$s^*(v, \Delta v) = s_0 + \max \left(0, vT + \frac{v\Delta v}{2\sqrt{ab}} \right) \quad (11.15)$$

has an equilibrium term $s_0 + vT$ and a *dynamical term* $v\Delta v/(2\sqrt{ab})$ that implements the “intelligent” braking strategy (see Sect. 11.3.4).⁷

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Intelligent Driver Model

We can easily interpret the model parameters by considering the following three standard situations:

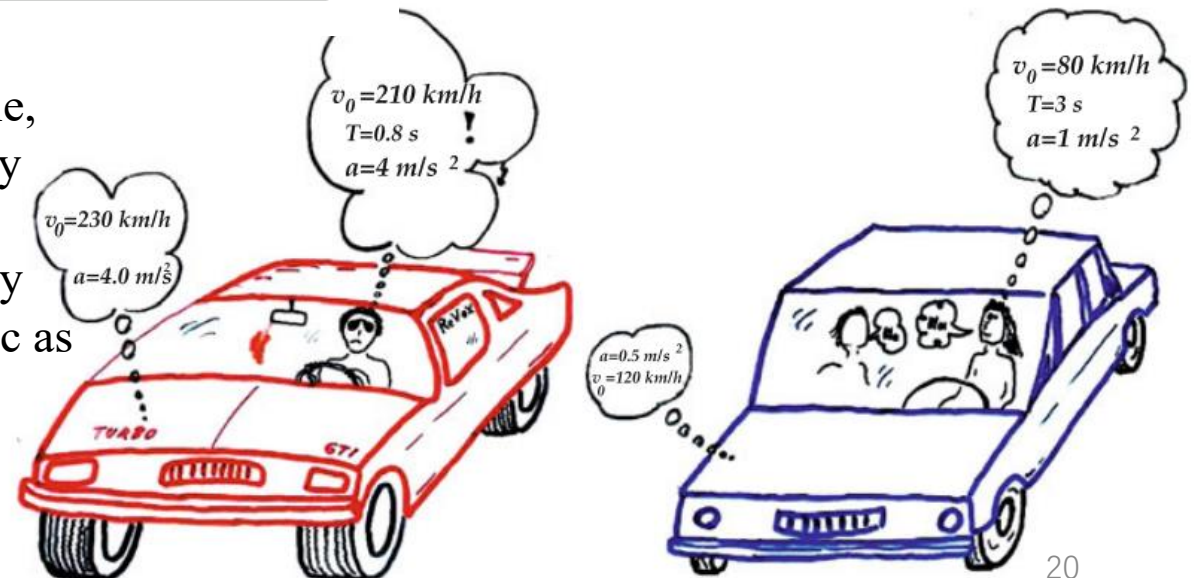
- When accelerating on a free road from a standstill, the vehicle starts with the maximum acceleration a .
- When following a leading vehicle, the distance gap is approximatively given by the safety distance $s_0 + vT$
- When approaching slower or stopped vehicles, the deceleration usually does not exceed the comfortable deceleration b .

Intelligent Driver Model

Table 11.2 Model parameters of the Intelligent Driver Model (IDM) and typical values in different scenarios (vehicle length 5 m unless stated otherwise)

Parameter	Typical value	Typical value
	Highway	City traffic
Desired speed v_0	120 km/h	54 km/h
Time gap T	1.0 s	1.0 s
Minimum gap s_0	2 m	2 m
Acceleration exponent δ	4	4
Acceleration a	1.0 m/s ²	1.0 m/s ²
Comfortable deceleration b	1.5 m/s ²	1.5 m/s ²

Each parameter describes a well-defined property . For example, transitions between highway and city traffic, can be modeled by solely changing the desired speed. All other parameters can be kept constant, modeling that somebody who drives aggressively (or defensively) on a highway presumably does so in city traffic as well.



Intelligent Driver Model

$$\dot{v} = a \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right] \text{ IDM.} \quad (11.14)$$

$$s^*(v, \Delta v) = s_0 + \max \left(0, vT + \frac{v\Delta v}{2\sqrt{ab}} \right) \quad (11.15)$$

The term $v\Delta v/(2\sqrt{ab})$ in the desired distance s^* (11.15) of the IDM models the dynamical behavior when approaching the leading vehicle. The equilibrium terms $s_0 + vT$ always affect s^* due to the required continuous transitions from and to the equilibrium state. Nevertheless, to study the braking strategy itself, we will set these terms to zero, together with the free acceleration term $a[1 - (v/v_0)^\delta]$ of the IDM acceleration equation. When approaching a standing vehicle or a red traffic light ($\Delta v = v$), we then find

$$\dot{v} = -a \left(\frac{s^*}{s} \right)^2 = -\frac{av^2(\Delta v)^2}{4abs^2} = -\left(\frac{v^2}{2s} \right)^2 \frac{1}{b}. \quad (11.16)$$

Intelligent Driver Model

With the *kinematic deceleration* defined as

$$b_{\text{kin}} = \frac{v^2}{2s}, \quad (11.17)$$

this part of the acceleration can be written as

$$\dot{v} = -\frac{b_{\text{kin}}^2}{b}. \quad (11.18)$$

When braking with deceleration b_{kin} , the braking distance is exactly the distance to the leading vehicle, thus b_{kin} is the minimum deceleration required for preventing a collision. With Eq. (11.18), we now understand the self-regulating braking strategy of the IDM:

$b_{\text{kin}} > b$
 e.g. $b_{\text{kin}} = 2 \text{ m/s}^2$ Driver's hobby is 1 m/s^2 : $|\dot{v}| = \frac{2^2}{1} = 4 \text{ m/s}^2$

$b_{\text{kin}} < b$
 e.g. $b_{\text{kin}} = 0.5 \text{ m/s}^2$ Driver's hobby is 2 m/s^2 $|\dot{v}| = \frac{0.5^2}{2} = 0.125 \text{ m/s}^2$

Intelligent Driver Model

Hence, the braking strategy is *dynamically self-regulating* towards a situation in which the kinematic deceleration equals the comfortable deceleration. One can show (see Problem 11.4) that this self-regulation is explicitly given by the differential equation

$$\begin{aligned}
 b_{kin} &= \frac{v^2}{2s} \\
 \frac{ds}{dt} &= -v \\
 \frac{dv}{dt} &= a = \dot{v} = -\frac{b_{kin}^2}{b}
 \end{aligned}$$

$$\begin{aligned}
 \frac{db_{kin}}{dt} &= \frac{d}{dt} \left(\frac{v^2}{2s} \right) = \frac{2v \frac{dv}{dt} 2s - v^2 2 \frac{ds}{dt}}{4s^2} \\
 &= \frac{2vs \frac{dv}{dt} - v^2 \frac{ds}{dt}}{2s^2} \\
 &= \frac{v}{s} \frac{dv}{dt} - \frac{v^2}{2s^2} \frac{ds}{dt} \\
 &= -\frac{vb_{kin}^2}{sb} + \frac{v^3}{2s^2} \\
 &= \frac{-vb_{kin}^2}{sb} + \frac{vb_{kin}}{s} \\
 &= \frac{vb_{kin}}{sb} (b - b_{kin})
 \end{aligned}$$